ECE1512 Project A Assignment Details

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Below we include the grade distribution for Project A.
Submission deadlines: Report and code: November 4, 2026.
Presentation materials: December 4, 2026.

Multi-Instance Learning (MIL) ([100 Points])

1. **Literature Review [20 Points]**
   You should provide a literature review on existing MIL methods. Ideally, you want to convey that you understand how it works and how we train/evaluate models for this task.

2. **Baseline Evaluation Summary [50 Points]**
   Using the provided ABMIL [1] training code, run the baseline model on CAMELYON16, CAMELYON17, and BRACS. Report results for each dataset. Prepare a concise report that includes:

   - Key observations about the model structure and training procedure.
   - The baseline performance metrics (e.g., accuracy, AUC, loss curves).
   - A brief explanation of what each model component (encoder, aggregator, classifier) contributes to the final prediction.

3. **Research Extensions [30 Points]**
   Propose **two** research ideas for improving or extending the ABMIL baseline.
   Implement and evaluate **one of the two ideas**. Possible directions include explainability, advanced model architectures, and alternative training objectives.

   - Two proposals (10 points): Clearly explain each idea, its motivation, and the rationale behind the design choices.
   - Implementation and evaluation of one selected idea (20 points): Implement and compare the selected extension with the ABMIL baseline on all three datasets and discuss the observed results.

   A negative result will not be penalized solely for failing to improve performance. Grading will focus on the soundness of the design, the quality of the experiments, and the justification and analysis provided.

What to submit & Evaluation

(a) You need to submit a written report (one per group) **in PDF format using the ECE1512 Quercus page submission utility.** Your report should be approximately **in IEEE style**[1] . List any additional references you have used in your work. You need to provide the code, data, figures and any other auxiliary material used in your work. To that end, your written report should include a link to your Project A GitHub repository. Your submission should include the baseline implementation, two research proposals, and the implementation and evaluation of one selected extension.

(b) **Project evaluation will include an examination of the code, notebooks, figures, and other auxiliary material posted on your GitHub repo.** It is highly recommended that you include a README document on your project's GitHub page.

(c) **Ouriginal "plagiarism" scores** will be visible upon (report) submission, so make sure that you are not penalized for plagiarism in uncited text and code portions (if applicable).

(d) Make sure your report submission is desigend to be used simultaneously with the Github-Repo. To do this, use **Machine Learning Paper Reproducibility Checklist** that clearly acknowledge that the report and the code are two separate artefacts, each with their own checklist.

(e) **Late submissions will not be accepted.**

References

[1] Maximilian Ilse, Jakub Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *International conference on machine learning*, pages 2127–2136. PMLR, 2018.

[1] `https://www.ieee.org/conferences/publishing/templates`